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Table 1 **Ethical issues, hazards and risks**

Ethical issue	Ethical hazard	Ethical risk	Mitigation	Comment	Verification/ Validation
Societal	Loss of trust (human robot)	Robot no longer used or is misused, abused	Design to ensure reliability in behaviour	If unexpected behaviour occurs, ensure traceability to help explain what happened	User validation
	Deception (intentional or unintentional)	Confusion, unintended (perhaps delayed) consequences, eventual loss of trust	Avoid deception due to the behaviour and/or appearance of the robot and ensure transparency of robotic nature	–	Software verification; user validation; expert guidance
	Anthropo-morphization	Misinterpretation	Avoid unnecessary anthropomorphization Clarification of intent to simulate human or not, or intended or expected behaviour	See deception (above) Use anthropomorphization only for well-defined, limited and socially-accepted purposes	User validation; expert guidance
	Privacy and confidentiality	Unauthorized access, collection and/or distribution of data, e.g. coming into the public domain or to unauthorized, unwarranted entities	Clarity of function Control of data, justification of data collection and distribution Ensure user awareness of data management and obtain informed consent in appropriate contexts	Privacy by design Data encryption, storage location, adherence to legislation	Software verification
	Lack of respect for cultural diversity and pluralism	Loss of trust in the device, embarrassment, shame, offence	Awareness of cultural norms incorporated into programming	Organizational, professional, regional	Software verification; user validation
	Robot addiction	Loss of human capability, dependency, reduction in willingness to engage with others, isolation	Raise awareness of dependency	A difficult area, particularly in relation to vulnerable people Careful evaluation of potential applications is needed	User validation; expert guidance

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Societal	Employment	Social dislocation, job replacement, unemployment, loss of skills/need to re-skill	Targeted deployment of robots to unsocial, hazardous activities, organizational; design	–	Economic and social assessment
Application	Misuse	All in this column	Minimize potential for unethical use via ethical risk assessment Adherence to standards and regulations	Innovate responsibly	Software verification
	Unsuitable divergent use	Unsuitable or inappropriate outcomes	Consider balance between useful/positive divergent use and inappropriate usage, inherent design	–	Software verification
	Dehumanization of humans in the relationship with robots	Inappropriate control exercised by the robot Loss of respect for human dignity, dehumanization and human rights	Inbuilt limits, design and implementation principles and practice	–	User validation; expert guidance
	Inappropriate "trust" of a human by robot	Malign or inadequate human control	Model of appropriate human control	Design robot to reach safe state with respect to any other tasks the robot is executing	Software verification; user validation
	Self-learning system exceeding its remit	Inappropriate use of resources	Limits on extent of self-learning without human monitoring	Keeping the human in the loop is usually advisable	Software verification
Commercial/ Financial	Approbation of legal responsibility and authority	Failure to meet fair contract conditions leading to illegality, inappropriate actions, avoidance of responsibilities	Attention to contracts, recognizing the characteristics of the robot, its intended domain of application and its operators	An issue involving corporate social responsibility, entraining legal and social considerations	Software verification; expert guidance

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Commercial/ Financial	Employment issues	Job replacement, job change, unemployment, loss of tax revenue	Appropriate support networks, appropriate taxation, retraining opportunities	Note literature on robot economics	Economic and social assessment
	Equality of access	Propagation of the "digital divide", isolation of minorities, non-compliance with human rights legislation	Inclusive design of robot behaviour to conform with Corporate Social Responsibility, and recognition of characteristics of intended application domain Support networks to minimize risks	–	Legal assessment; software verification
	Learning by robots that have some degree of behavioural autonomy	Robot might develop new or amended action plans, or omit steps in processes, that could have unforeseen consequences for safety and/or quality of outcomes	Designers can enable robots to inform their operators when a new form of behaviour has been developed	Important consideration for human-robot trust	Software verification; user validation
	Informed consent	Unaware operators causing accidents, unwanted consequences, unfair and inequitable responsibilities placed upon consenter, inability to respond to situations	Design contracts to make explicit, in plain language, what the consenter is agreeing to, as far as the robot is concerned, including risks Also, contracts to make plain what personal or private information could be passed on to the network(s) to which the robot is connected	In principle, the actions of sets of robots, especially when networked, is nearly unlimited There are privacy and security issues that might arise as robots working in close co-ordination with humans become more informed about human behaviour	User validation

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Commercial/ Financial	Informed command	User is unaware of extent or legality or social acceptance of the tasks given to the robot, consequences of the tasks might ramify in unexpected ways and extents	As a principle, robots ought to only act on the basis of a properly-constructed command. If the robot cannot construe such a command from the human's communication, it ought to seek further guidance	A generally accepted principle is that "the last person to issue a command is responsible for its outcome". For equity, safety and security, it is necessary to ensure that operators are as informed as possible about the capabilities and limitations of the robots that they utilize [A)]	Software verification
Environmental	Environmental awareness (robot and appliances)	Cause concerns about wastage and destruction of the environment Failure to conform to regulations and/or codes of practice designed to protect the environment resulting in harm to the society	Adopt good/best management practice in design (see BS EN ISO 9000 series) and support for recycling and other circular economy activities	Likely to be of steadily-increasing importance	Lifecycle compliance tests
	Environmental awareness (operations and applications)	Execution of non-sustainable actions, harm to local situation, reputational harm	Good/best management practice (see BS EN ISO 14000 series) built into contract terms, design of robot sensing to ensure recognition and labelling of physical environment, and, particularly in relation to pick-and-place actions, sources and sinks are known	–	Compliance tests

[A)] For examples of "properly-constructed commands", designers and operators can consult NATO's C-BML (Coalition Battle Management Language). See **5.1.12**.